

Modeling of xVA Indicators for Counterparty Risk Management

This project focuses on the modeling of xVA indicators (CVA, DVA, FVA, MVA and KVA), which are widely used in financial institutions to account for counterparty risk and various economic costs in the valuation of derivative products.

The objective is to develop a Python prototype capable of simulating risk factors, estimating future exposures and computing these valuation adjustments in a simplified but realistic framework.

Team Number : 491

Project Partner : Charaf Louhmedi - External lecturer at ESILV

Project Team : Léa NAPASEUTH, Mathilde DESTAILLEUR, Chiara GODET, Chloé CHAUFOR, Emilien COMBARET, Théo DURAND

Team Member	Role
Léa NAPASEUTH	Project coordination, monitoring of project progress, preparation of the closure report and contribution to the documentation of the model
Mathilde DESTAILLEUR	Project documentation and soft skills report, contribution to testing, verification of results and project coordination tasks
Chiara GODET	Design of the project poster and communication materials, participation in the interpretation and presentation of results
Chloé CHAUFOR	Preparation of the technico-functional document, contribution to the theoretical analysis of counterparty risk and xVA indicators
Emilien COMBARET	Development of the visualization interface and dashboard, contribution to the implementation of the Python prototype and preparation of the final presentation
Théo DURAND	Development of the Python simulation framework (Monte Carlo simulations and xVA calculations), participation in model validation and numerical testing

Table of Contents

1 Objectives Set versus Results Achieved	3
1.1 Project Initial targets	3
1.2 Results Achieved	3
1.3 List of Deliverables.....	5
2 Methodological reviews	6
3 Risk Management	7
4 Constraints	7
5 Technical Review.....	8
6 Engineering standards	10
7 “Post-project” tasks	11
8 Project Completion Recommendations	11
9 Annexes to the closure report	12

1 Objectives Set versus Results Achieved

1.1 Project Initial targets

The main objective of the project was to understand and model the main xVA indicators used by financial institutions to account for counterparty risk and other economic costs in derivative valuation.

More specifically, the project aimed to :

- Understand the theoretical framework of counterparty risk and valuation adjustments (CVA, DVA, FVA, MVA and KVA).
- Study the regulatory context, particularly the Basel III and Basel IV reforms.
- Simulate the evolution of financial risk factors such as interest rates and credit risk.
- Estimate the future exposure of a portfolio of derivatives using Monte Carlo simulations.
- Implement a Python prototype capable of computing the main xVA indicators.
- Provide a pedagogical tool illustrating the impact of these adjustments on the economic value of derivatives.

The project was also expected to deliver a clear and structured code architecture, allowing future improvements or extensions of the model.

1.2 Results Achieved

At the current stage of the project, most of the main objectives have been reached.

On the theoretical side, the team developed a solid understanding of counterparty risk, valuation adjustments and the post-crisis regulatory context. A full technico-functional document was produced in order to define the scope of the project, the modeling assumptions and the main implementation choices. This document also helped us clarify the links between the theoretical xVA framework and the quantitative approach used in the prototype.

On the technical side, a working Python prototype was developed. The model is based on Monte Carlo simulations of interest-rate scenarios in order to estimate future exposures and compute the main valuation adjustments. In the current version, the simulation framework relies on two correlated short-rate dynamics inspired by a multi-factor Hull–White setting, used to represent the €STR discounting curve and the Euribor 6M projection curve.

The prototype makes it possible to:

- simulate interest-rate paths over a 10-year horizon,
- revalue several rate products under simulated scenarios,
- compute positive and negative expected exposures,
- calculate simplified xVA metrics such as CVA, DVA, FVA and KVA.

The portfolio considered in the model includes several interest-rate instruments such as an IRS, a zero-coupon bond, a FRA, a cap and a swaption, which gives a more complete view of the exposure profiles generated by different products.

A major recent development of the project was also the creation of an interactive dashboard. This dashboard allows the user to select a financial instrument, modify key parameters such as notional, rate or volatility, launch the calculation through a local API, and visualize the results through interactive charts. This part of the project made the quantitative model more accessible and more pedagogical.

An additional validation step was carried out through the creation of an Excel support file used to verify and confirm the numerical results displayed in the dashboard. This file made it possible to compare the computed values and ensure the consistency of the outputs shown in the interface.

However, some parts are still being finalized. In particular, the code can still be improved in terms of robustness and structure, the visualization interface is still being refined, and some extensions of the xVA framework remain possible. Despite these remaining tasks, the project already provides a functional and coherent prototype illustrating the main mechanisms behind xVA calculations.

1.3 List of Deliverables

The project includes several deliverables:

Name of the file	Deliverables	Description
ClosureReport_PA4_25_491.pdf	Closure report	key descriptive information about the project
Code_pi2_xva	Python prototype/Dashboard	Implemente the simulation, of risk factors, exposure calculation and xVA computation
Dashboard validation	Excel file of calculation	Excel file used to verify the consistency of the numerical results displayed in the dashboard
Equipe-491_Projet-169_Doc_technico_fonctionel_pi2_xVA.pdf	Technico-functional document	Describe the objectives, architecture and modeling choices of the project
ClosurePres_PA4_25_491.pdf	Powerpoint	PowerPoint of the oral presentation for the project defense
Video_PA4_25_491.mp4	Video	short presentation video that that summarize in few minutes the project with demonstration of the code
Visuel_PA4_25_491.jpg	Poster	Poster for the ESILV showroom presenting the main concepts and results of the project
ResumeFR_PA4_25_491.txt	Abstract in French	Project follow-up reports documenting the evoluto, of the work during the semester in french
AbstractEN_PA4_25_491.txt	Abstract in English	Project follow-up reports documenting the evoluto, of the work during the semester in english

2 Methodological reviews

The project was carried out progressively, with a method that was both structured and flexible. At the beginning, the main priority was to understand the subject correctly, because xVA is not only a coding project but also a topic that requires a good financial and mathematical background. For this reason, the first phase of the work focused on bibliographical research, understanding the main valuation adjustments and identifying the regulatory context around counterparty risk.

Once the team had a clearer understanding of the subject, we moved to the definition of the scope of the model and the technical choices. This work led to the writing of the technico-functional document, which played an important role during the project because it helped us formalize the hypotheses, the portfolio composition, the simulation framework and the main implementation choices.

The development itself was done iteratively. We did not try to build the full project in one step. Instead, we advanced by layers: first the market framework and rate curves, then the simulation of the rate factors, then the valuation of the selected instruments, then the exposure profiles, and finally the simplified xVA calculations. This progressive method made the project easier to manage and allowed us to test each part before going further.

The team also adopted a collaborative way of working. Even if some members had a more visible role on certain tasks, the project was not divided into completely isolated blocks. Several members contributed to the theoretical work, documentation, testing, presentation material and discussions about implementation choices. This made the work more balanced and helped everyone understand the project globally.

Regular exchanges inside the team were important throughout the semester. We used meetings, shared documents and collaborative coding tools to follow the project, distribute tasks and react when some parts were taking more time than expected. Meetings with the tutor were also very useful because they helped us validate the direction of the project and refine some of the modeling choices.

Overall, the methodology used during the project was well adapted to the nature of the topic. It allowed us to manage a complex quantitative subject while still keeping a coherent progression from theory to implementation and then to visualization.

3 Risk Management

During the project, several risks were identified.

The first one was the theoretical complexity of the topic. xVA combines derivative pricing, counterparty risk, regulation and numerical methods, which makes the subject difficult to approach at the beginning. To reduce this risk, we spent significant time on the theoretical framework before moving to the implementation. The technico-functional document was especially useful in this phase because it helped us organize the ideas and avoid coding without a clear direction.

Another important risk concerned the availability of market data. In a real banking environment, xVA models are calibrated using professional market data sources. In our case, we did not have access to this kind of infrastructure, so we had to rely on a stylized but coherent market environment. This was a limitation, but it also allowed us to keep the project manageable in an academic context.

A technical risk was linked to the computational cost of Monte Carlo simulations. Since the project requires repeated valuation under many simulated scenarios, the code could quickly become slow or difficult to handle. To deal with this issue, we kept a reasonable number of scenarios and used numerical libraries such as NumPy and QuantLib to maintain acceptable computation times.

There was also a project management risk related to time. The project required not only technical work, but also several final deliverables such as the poster, the dashboard, the presentation, the video and the closure report. This risk was handled through internal coordination, task sharing and regular follow-up between team members.

In the end, these risks did not disappear completely, but they were managed well enough for the project to move forward in a consistent way.

4 Constraints

Several constraints influenced the development of the project.

The first one was the complexity of the subject itself. Modeling xVA is not limited to implementing formulas: it also requires understanding how future exposures are generated, how derivative products are revalued under simulated market conditions, and how these results are linked to credit and funding adjustments. This required a constant balance between theoretical accuracy and practical feasibility.

A second constraint was the lack of access to professional financial data. As a result, the project had to rely on simplified assumptions and stylized parameters. The model is therefore not intended to

reproduce a full banking production framework, but rather to provide a pedagogical and technically consistent prototype.

An additional practical issue was encountered when retrieving the Euribor 6M rate at the initial date t_0 . The initial website used for web scraping was no longer accessible, which required us to investigate several alternative data sources and retrieval methods.

The use of official APIs proved to be limited due to the lack of simple and free public access. Other web scraping sources, such as Investing.com and TradingView, were also considered but were not retained because of anti-bot mechanisms, unstable HTML structures and the risk of request blocking. An attempt to retrieve the data from the FRED database also resulted in repeated timeout issues, likely due to network restrictions and frequent request throttling.

After investigating these different approaches, we were able to resolve the Euribor 6M data retrieval issue by implementing a more suitable scraping approach. This allowed us to retrieve the required market data while maintaining a consistent and reproducible data collection process for the model.

As a fallback solution and modeling consistency check, we also considered approximating the Euribor 6M rate from the €STR rate using a spread:

$$\text{Euribor 6M} \approx \text{€STR} + \text{spread}$$

where the spread is typically considered to be in the range of 10 to 30 basis points. This approximation provided an alternative in case of temporary data-access issues and remained consistent with the multi-curve framework used in the project.

Time was also a significant constraint. The project had to be completed within the academic calendar, while several intermediate and final deliverables had to be prepared in parallel. This limited the possibility of developing every component to a fully industrial level.

Another constraint was computational. Even in a simplified framework, Monte Carlo simulation remains resource-intensive, especially when several products must be revalued repeatedly across multiple scenarios and time steps.

Finally, the project had to remain understandable and accessible. Since this is an academic project, it was important to avoid developing a model that would be unnecessarily complex to explain. The final framework therefore had to remain readable, reproducible and suitable for presentation and demonstration.

5 Technical Review

From a technical point of view, the project resulted in the development of a simplified but coherent framework for the calculation of xVA indicators on interest-rate products.

The quantitative core of the project was implemented in Python. The model relies on Monte Carlo simulations of interest-rate scenarios over a 10-year horizon, with a monthly time step and a fixed random seed in order to ensure reproducibility. In the current version, the simulation framework is based on two correlated short-rate dynamics inspired by a multi-factor Hull–White model. These two factors are used to represent the €STR discounting curve and the Euribor 6M projection curve.

Due to practical data availability issues, the Euribor 6M curve used in the model is not directly calibrated from market data but approximated from the €STR curve using a constant spread. This simplifying assumption is consistent with the academic scope of the project and reflects the typical relationship observed between OIS and Euribor rates in practice.

Using these simulated scenarios, several instruments are revalued at each date of the time grid. The products considered in the project include an IRS, a zero-coupon bond, a FRA, a cap and a swaption. This allows the model to generate different types of exposure profiles, including both linear and optional products.

Based on these revaluations, the project computes positive and negative expected exposures. In the current implementation, these exposures are calculated instrument by instrument rather than through a complete aggregation of the portfolio. This choice was sufficient for the pedagogical objective of the project, even if a full portfolio aggregation would be a natural extension of the work.

The xVA calculations are then built on top of these exposures. In the current model, the default probabilities used for CVA and DVA are introduced exogenously in a simplified way. FVA and KVA are also implemented through simplified approximations consistent with the academic scope of the project. MVA is presented in the framework but remains zero in the non-collateralized version of the current implementation.

Another important technical result of the project is the dashboard developed at the end of the project. This dashboard adds a front-end / back-end layer to the quantitative model and makes the project much easier to use and demonstrate. The front-end was developed in React and allows the user to:

- select a financial product,
- modify product parameters such as notional, rate or volatility,
- send a request to a local backend API,
- visualize the returned results through dynamic charts.

The dashboard was designed to improve the readability of the model outputs and to make the project more interactive. The use of interactive charts allows the user to observe the evolution of the computed metrics more clearly. Particular attention was also given to usability, with loading states, error handling and a clean organization of inputs and outputs.

In order to strengthen the reliability of the interface, an Excel validation file was also created to cross-check some of the numerical results returned by the dashboard.

Even though the final framework remains simplified compared with real xVA systems used in financial institutions, it succeeds in combining three important dimensions of the project: quantitative modeling, Python implementation, and user-oriented visualization.

6 Engineering standards

Several engineering and methodological principles were followed during the project in order to keep the work clear, reproducible and technically consistent.

First, the project follows common good practices in scientific programming. Even if the current Python implementation is still grouped mainly in a principal script, the calculation chain was built in a structured way, with clearly identified stages: data initialization, market curve construction, rate simulation, product valuation, exposure calculation and xVA computation. This makes the workflow understandable and easier to improve later.

Second, the project relies on standard numerical methods used in quantitative finance. Monte Carlo simulation was used to estimate future exposures, and numerical tools such as NumPy and QuantLib were used to implement the simulation and valuation framework.

Third, the project follows standard financial modeling ideas commonly used in the industry. In particular, it distinguishes discounting and projection curves, uses simulated short-rate dynamics, and links future exposures to xVA metrics through simplified but meaningful formulas.

Finally, the whole project is consistent with the broader regulatory logic of Basel III and Basel finalization, since it focuses on the economic and prudential impact of counterparty risk on derivative valuation

7 “Post-project” tasks

Although the main technical core of the project is already functional, several post-project tasks remain.

The first task is to further stabilize and clean the Python implementation. Some parts of the script can still be improved in terms of readability, robustness and organization.

A second important task is to finalize the dashboard. The interface is already operational, but it can still be improved with additional visualizations, cleaner parameter handling and a stronger connection between the quantitative backend and the displayed outputs.

Another possible continuation of the project would be to move from instrument-by-instrument exposures to a true portfolio-level aggregation. This would make the framework more realistic from a risk-management perspective.

The project could also be extended by introducing a collateralized framework, dynamic credit modeling, or more realistic market calibration. These extensions were identified during the project but could not be fully implemented within the available time.

Finally, the last remaining tasks concern the final deliverables: the polishing of the presentation slides, the showroom poster, the video and the final project communication materials.

8 Project Completion Recommendations

This project highlighted several useful recommendations for future work.

First, on a subject like xVA, it is essential to spend enough time at the beginning on theory. Without a clear understanding of the financial meaning of CVA, DVA, FVA or KVA, it is very easy to lose time during the implementation phase.

Second, it is important to define early what the objective of the project really is. In our case, trying to build a fully realistic banking model would have been too ambitious. What worked better was to aim for a simplified but coherent framework and then improve it progressively.

Another recommendation would be to connect the quantitative work to a visualization tool as early as possible. The dashboard significantly improved the way the project could be presented and understood. It made the results more concrete and gave the project a more complete dimension.

It would also be useful in a future version to improve the code structure earlier in the process, so that new extensions can be added more easily.

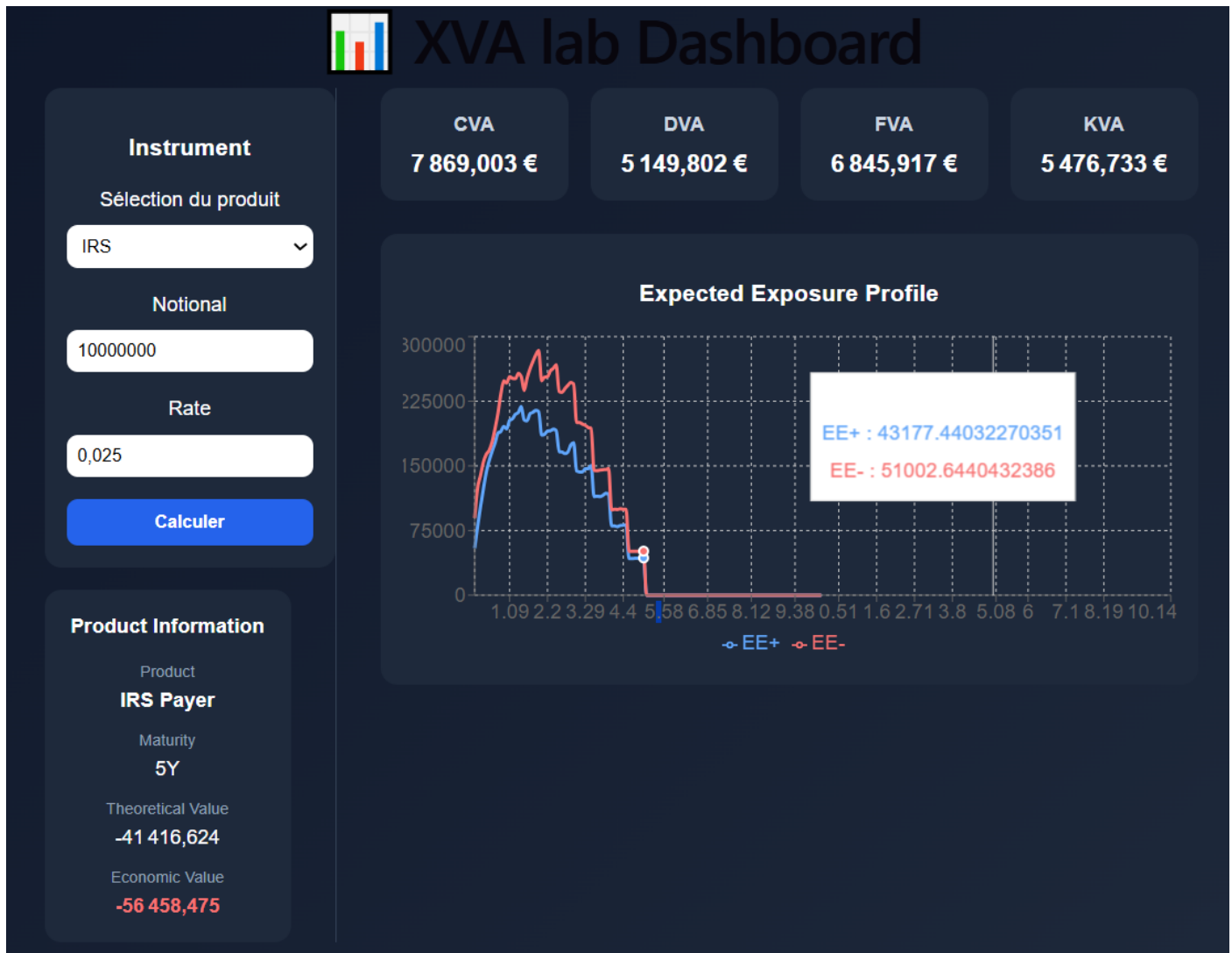
Finally, having access to more realistic market data would clearly improve the quality of the results and the credibility of the calibration. Even if our simplified framework was enough for an academic project, this remains one of the main directions for improvement.

Overall, the project was very useful because it combined finance, programming, modeling and teamwork, and it gave us a better understanding of how quantitative tools can be used to study counterparty risk.

9 Annexes to the closure report

Link of the Dashboard : <https://xva-dashboard.vercel.app/>

Dashboard :



Poster

Équipe 491

Modeling xVA Adjustments

Valuation of Derivative Products with Risk Adjustments

Chiara G., Mathilde D., Emilien C., Chloé C., Léa N., Théo D.

Context	Methodology	Results
Since 2008, derivative valuation must incorporate: • Counterparty Risk • Funding • Regulatory Capital	↓ Rates (HW) ↓ Monte Carlo ↓ Exposure $EE(t)$ ↓ xVA ↓ Economic Portfolio Valuation ↓ Concepts CVA Counterparty Default DVA Own Default FVA Funding MVA Margin KVA Capital Value = $V_{th} - CVA - DVA - FVA - MVA - KVA$ xVA are essential in banking for managing risk and pricing.	Simulation of Rates Expected Exposure $EE(t)$
Problem	Implementation	Limits
How to value a derivative while accounting for risks?	• Python • NumPy • Vectorization	⚠ Exogenous credit risk ⚠ No collateral (CSA) ⚠ No wrong-way risk ⚠ Monte Carlo computational cost
Objectives		
✓ Model xVA ✓ Simulate risks ✓ Compute exposures ✓ Develop a Python prototype		
Conclusion This project demonstrates how quantitative modeling and Monte Carlo simulation can be combined to accurately integrate counterparty risk into the economic valuation of derivatives.		

Team logo :

